

Cybersecurity Practice Exam Questions

Open-Ended Questions

1. Define Open-Source Intelligence (OSINT) and describe how it can be used during a cybersecurity investigation.
2. What is passive reconnaissance? Explain its purpose and list three publicly accessible sources that can be used to gather information.
3. Compare and contrast asymmetric and symmetric cryptography. Provide one practical use case for each.
4. Define the term 'cryptography'. Why is it essential in the context of modern digital communication?
5. Explain the difference between encryption and decryption. Include an example of how both processes are used in securing a message.
6. What is a vulnerability in cybersecurity? How does a zero-day vulnerability differ from a known vulnerability?
7. Describe the cyber kill chain model. Outline its stages and explain how it is used in cybersecurity defense strategies.
8. What is SSL (Secure Sockets Layer)? Describe its function and explain how it protects communication over the internet.

Multiple-Choice Questions

1. Where is a Tier-1 Internet Service Provider (ISP) typically located?
 - A. At a local data center
 - B. Within a national telecommunications office
 - C. At the core of the global internet infrastructure
 - D. In residential areas
2. In which OSI layer does a router primarily operate?
 - A. Physical Layer

- B. Network Layer
- C. Transport Layer
- D. Data Link Layer

3. Which of the following is an asymmetric encryption algorithm?

- A. AES
- B. DES
- C. RSA
- D. Blowfish

4. Which standard protocol is primarily used for substation data modeling and communication in smart grids?

- A. IEC 61236
- B. IEC 61850
- C. IEEE 802.3
- D. OPC-UA

5. Which of the following is NOT considered a type of spyware?

- A. Keylogger
- B. Trojan
- C. Virus
- D. Remote Access Tool (RAT)

6. What was the final stage of the 2015 cyberattack on the Ukrainian power grid?

- A. Delivery of malware via email
- B. Reconnaissance of network topology
- C. Wiping systems and cutting off UPS and backup systems
- D. Data exfiltration to external servers

7. How is a wide area network (WAN) typically classified in terms of trust?

- A. Fully trusted

- B. Semi-trusted
- C. Untrusted
- D. Trusted if encrypted

8. What is the primary goal of host discovery in a network scan?

- A. To capture packets from a target
- B. To identify services running on a port
- C. To determine which devices are active on the network
- D. To exploit vulnerable hosts

9. In Zenmap (Nmap GUI), what does a port status of 'filtered' typically indicate?

- A. The port is confirmed closed
- B. The port is open but no service is running
- C. The probe was blocked or dropped, possibly by a firewall
- D. The port has been spoofed by the target